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Domain: Data Science

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Karachi AQI Forecasting System

Project Objective

To develop a real-time AQI prediction system that fetches raw pollutant and weather data, computes engineered features, trains multiple ML models, and provides forecasted air quality insights via an interactive dashboard. The system is built with a scalable and automated ML pipeline and provides hourly updates, explainability via SHAP, and hazard alerts.

Project Overview

Air pollution is one of the most pressing environmental challenges in urban areas, significantly affecting public health, climate, and the overall quality of life. In cities like Karachi, rapid industrialization, traffic congestion, and limited environmental regulation have led to deteriorating air quality. The Air Quality Index (AQI) serves as a standardized measure to assess and communicate the level of air pollution in a region. Real-time AQI prediction plays a vital role in empowering authorities and citizens to make timely decisions, reduce exposure to harmful pollutants, and plan interventions effectively.

This project aims to develop an end-to-end, scalable AQI forecasting system using real-time data ingestion, machine learning models, and automation tools. The system fetches weather and pollutant data from external APIs such as AQICN, IQAir, and OpenWeather, processes this raw data into engineered features, and uses these features to train and evaluate forecasting models. These models are capable of predicting current and future AQI levels with the objective of issuing alerts when hazardous conditions are detected.

Since AQI prediction produces continuous numerical values rather than discrete categories, this is framed as a **regression problem**. It falls under **supervised learning** because the models are trained on historical labeled data, where the input features (e.g., temperature, wind speed, pollutant concentrations) are mapped to known AQI values. Various regression algorithms are evaluated—such as Linear Regression, Ridge Regression, and XGBoost—and the final model is selected based on regression-specific evaluation metrics rather than classification metrics. Accuracy is meaningful in classification tasks because predictions are discrete labels that can be compared directly to the true class. In regression, predictions are

continuous values (e.g., AQI = 157.32), and an “exact match” is almost never possible. Instead of measuring the percentage of exactly correct predictions, regression focuses on how close the predicted values are to the actual values. Therefore, metrics like **Mean Absolute Error (MAE)**, **Mean Squared Error (MSE)**, **Root Mean Squared Error (RMSE)**, and **R^2 score** are used to assess performance. These metrics quantify the average magnitude of errors, penalize large deviations, and measure the proportion of variance explained by the model, which is more appropriate for continuous outputs.

The project pipeline integrates essential components of a modern ML system, including feature engineering, model training and evaluation, a feature store, model registry, and CI/CD pipelines using tools like GitHub Actions or Apache Airflow. Predictions are served via a FastAPI or Flask backend and visualized through an interactive Streamlit or Gradio dashboard, offering users a real-time view of AQI levels and forecasts.

Additionally, the system leverages **SHAP** for model explainability, ensuring transparency in prediction decisions. An alert mechanism is incorporated to notify users of various AQI levels, enabling them to take proactive health and safety measures.

This comprehensive solution demonstrates how machine learning, automation, and cloud technologies can be combined to address critical urban challenges like air pollution, ultimately contributing to public awareness and environmental monitoring.

Project Steps

The Data Science Steps are as follows:

- Data Collection

Real-time weather and air pollutant data were collected using external APIs such as **AQICN, IQAir** and **OpenWeather**.

- Data Preprocessing

The collected data was cleaned by handling missing values, removing unnecessary columns (like forecast fields), and formatting the time-related fields for consistency.

- Feature Engineering

Meaningful features were created to improve model performance, including:

- Time-based features such as hour, day, and month
- Derived features such as AQI change rate, pollutant ratios, rolling averages, and weather features like temperature and humidity.

- Exploratory Data Analysis (EDA)

  The data was analyzed to identify trends, seasonal patterns, distributions, and correlations between pollutants and AQI values.

- Modeling

  Several machine learning and deep learning models were trained and evaluated, including:
  - **Scikit-learn models**: XGBoost , Linear Regression, Ridge Regression

  The models were validated using **TimeSeriesSplit**, and evaluated using metrics such as **MAE**, **RMSE**, and **R^2 score**.

- Model Deployment

  The best-performing model was saved in a model registry and deployed through an API using **FastAPI**.

- CI/CD Automation

  Continuous integration and deployment pipelines were created using **GitHub Actions**, enabling:

  - Hourly execution of the feature engineering script

- Daily execution of the model retraining script

- Model Explainability & Alerts

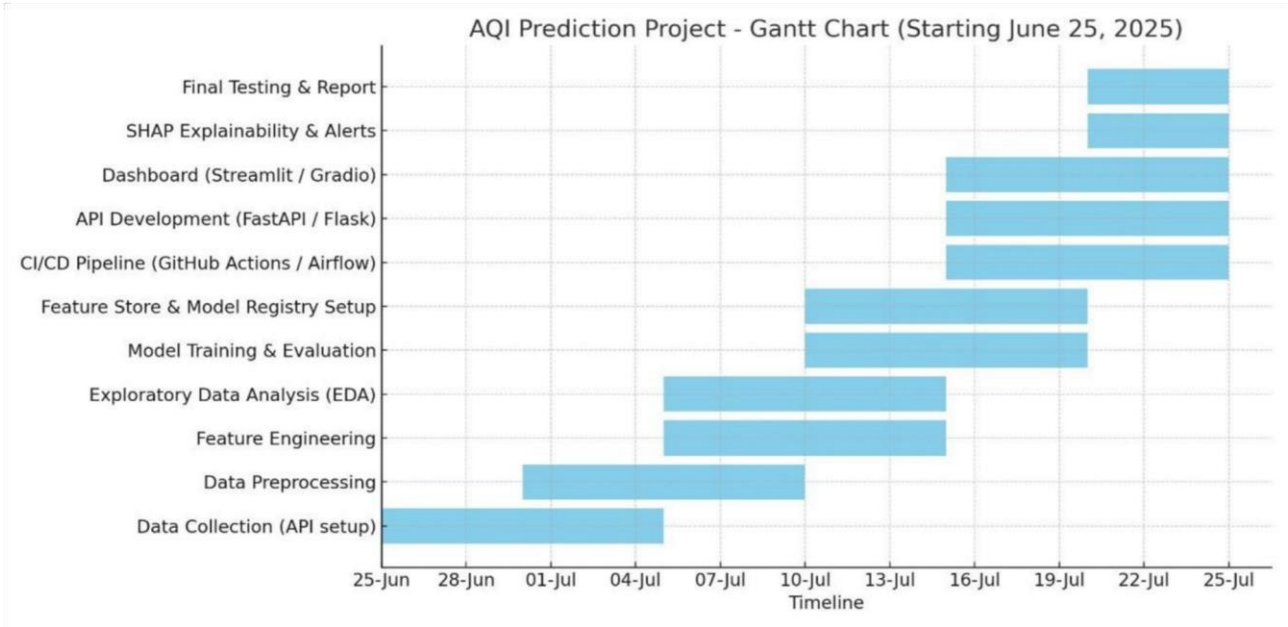
  SHAP was used to explain model predictions and understand feature importance. An **alert mechanism** was added to notify users when AQI levels exceed hazardous thresholds.

- Interactive Dashboard

  A real-time dashboard was built using **Gradio** to display:

  - Current AQI
  - 3-day AQI forecast
  - Alert levels and health-related advisories

Project Timeline



Description of each step

1. Data Collection

This step involves gathering real-time weather and air quality data of Karachi from external APIs such as AQICN and OpenWeather. The collected data includes pollutants (e.g., PM2.5, PM10, CO, NO_2, etc.) and meteorological parameters (e.g., temperature, humidity, pressure).

2. Data Preprocessing

In this stage, the raw data is cleaned and prepared for modeling. This includes handling missing values, removing irrelevant columns, formatting timestamps, and ensuring data consistency. Additional steps include visualizing null values using heatmaps to identify data gaps, and retrieving all column names to understand the dataset structure before processing begins.

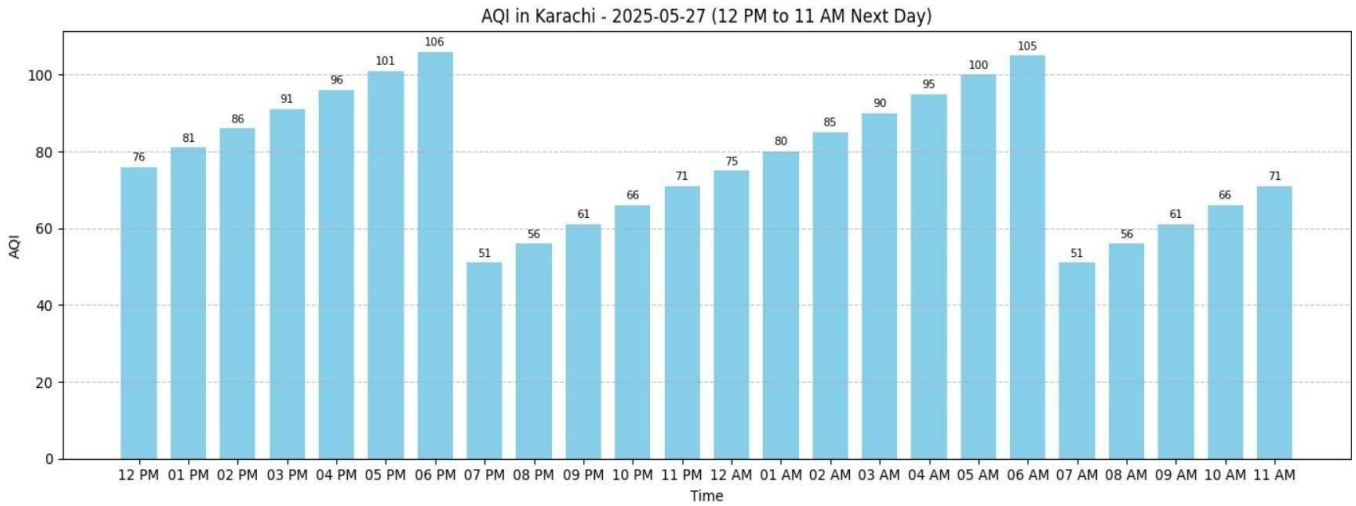
3. Feature Engineering

New features are created to enhance model performance. These include timebased features (hour, day, month), derived features (AQI change rate, pollutant ratios), and weather features (temperature, humidity). These engineered features serve as inputs to the prediction models.

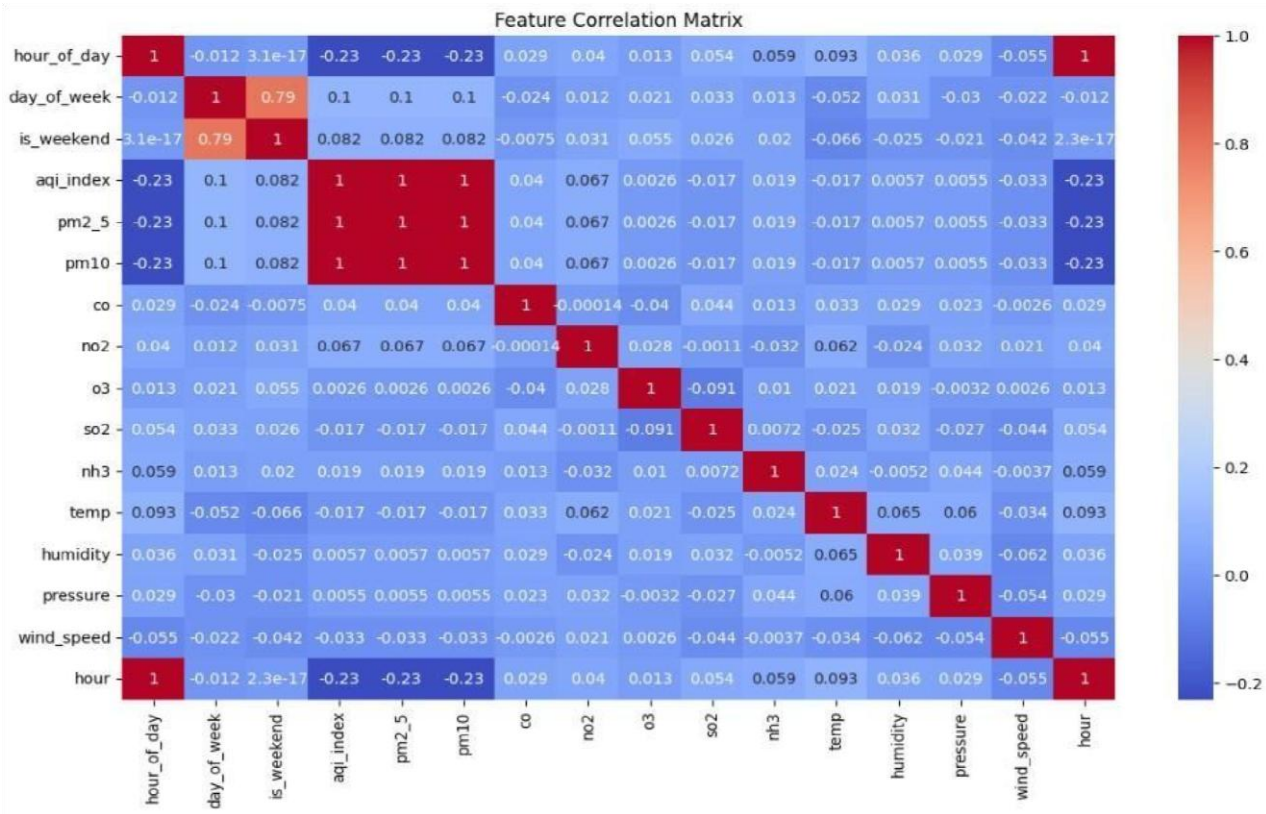
4. Exploratory Data Analysis (EDA)

EDA is used to understand the underlying patterns and relationships in the data. Visualizations such as time-series plots, correlation heatmaps , and distribution graphs are used to gain insights into pollutant behavior and AQI trends.

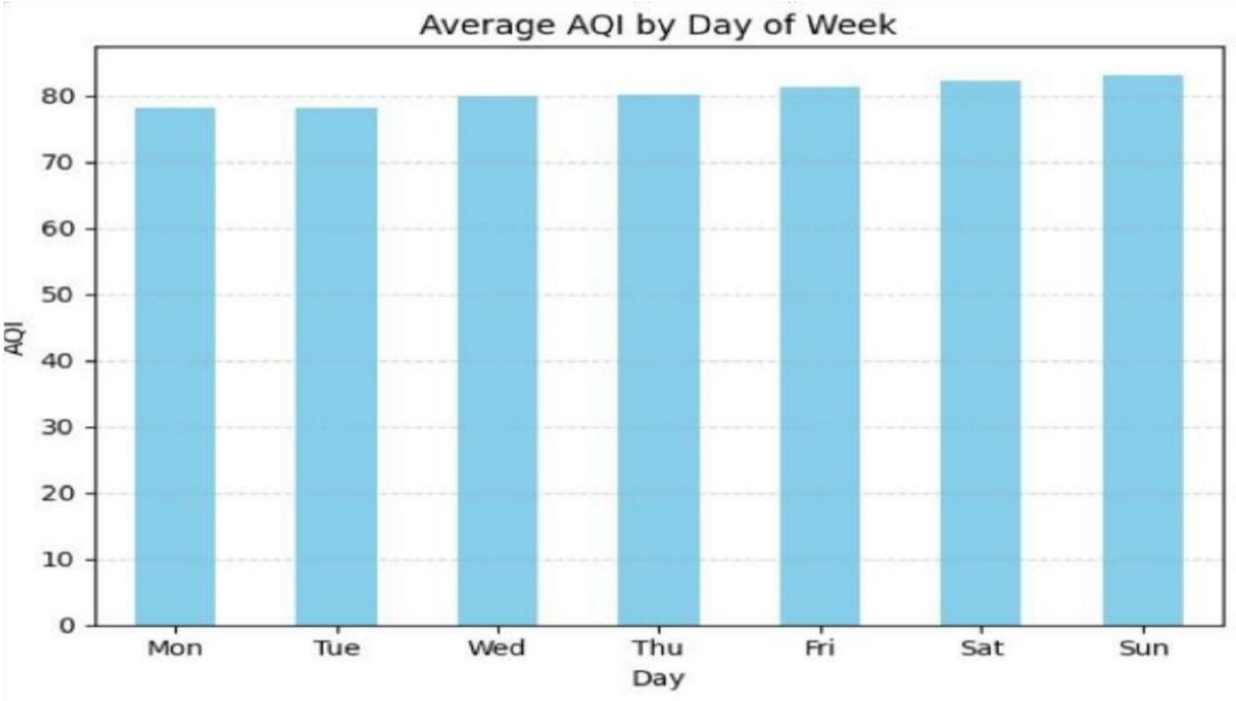
Time - Series Plot Outputs



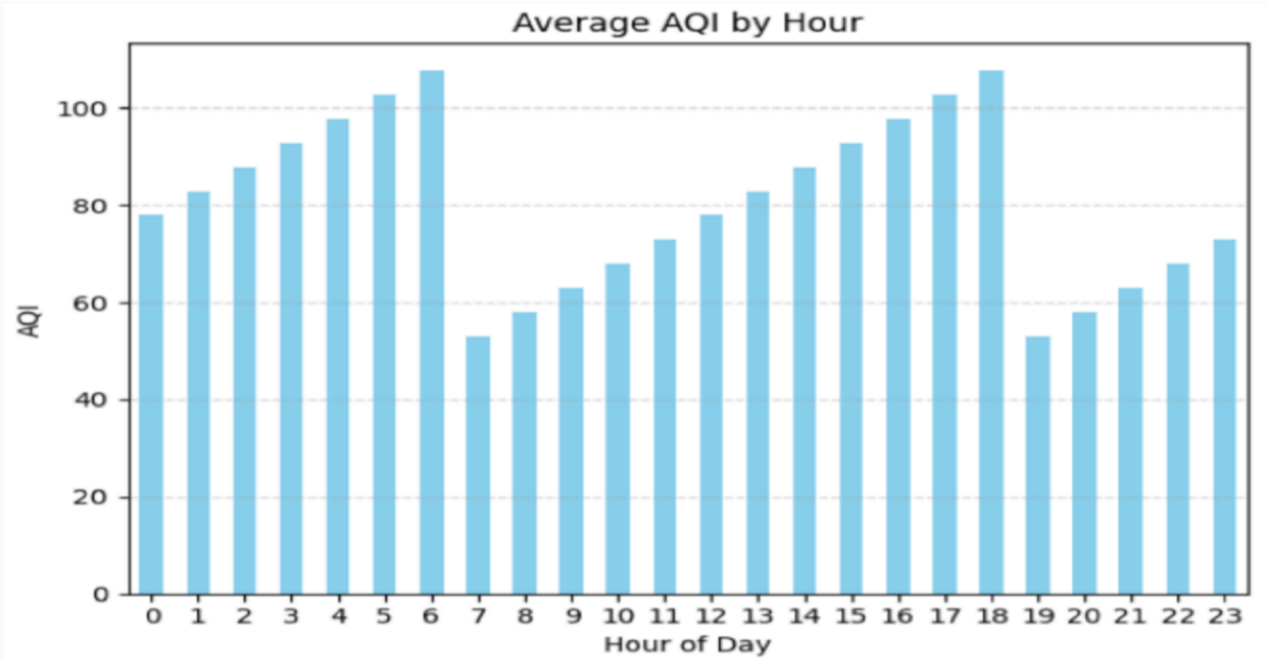
Correlation Heatmap



Average AQI by Day of Week



Average AQI by Hours of Day



5. Model Training & Evaluation

Various machine learning and deep learning models are trained to predict AQI. Models like Linear Regression, Ridge Regression, Random Forest are evaluated using metrics like MAE, RMSE, and R^2. Time Series Split is used to maintain temporal integrity.

Result After each Model Training & Evaluation

For AQI forecasting, the system was trained using supervised regression models, as the target variable (AQI value) is continuous rather than categorical. The training dataset consisted of historical pollutant and weather data obtained from APIs such as AQICN, IQAir, and OpenWeather, preprocessed into suitable features for model learning.
A variety of regression algorithms were tested, including **Linear Regression, Ridge Regression, and XGBoost**. The models were evaluated using **regression-specific metrics** rather than classification accuracy, as exact label matching is not meaningful for continuous predictions.

To check model accuracy in regression tasks, the following thresholds were applied:

1. **MAE or RMSE between 0 and 0.5** → Very small errors — *Excellent prediction*
2. **MAE or RMSE between 0.5 and 1.0** → Small errors — *Great prediction*
3. **MAE or RMSE between 1.0 and 3.0** → Moderate errors — *Good prediction*
4. **MAE or RMSE between 3.0 and 5.0** → Noticeable errors — *Fair prediction*
5. **MAE or RMSE greater than 5.0** → Large errors — *Poor prediction*
6. **R^2 score close to 1 (0.9–1.0)** → Model explains almost all variance — *Excellent fit*
7. **R^2 score between 0.7 and 0.9** → *Good fit* — Strong prediction power
8. **R^2 score between 0.5 and 0.7** → *Moderate fit* — Fair prediction
9. **R^2 score below 0.5** → Weak fit — *Poor prediction quality*

By applying these evaluation thresholds, the most suitable model was selected based on the lowest error values (MAE, RMSE) and the highest R^2 score. This ensured that the final deployed model offered the most reliable AQI predictions for real-world use.

1. XGBoost

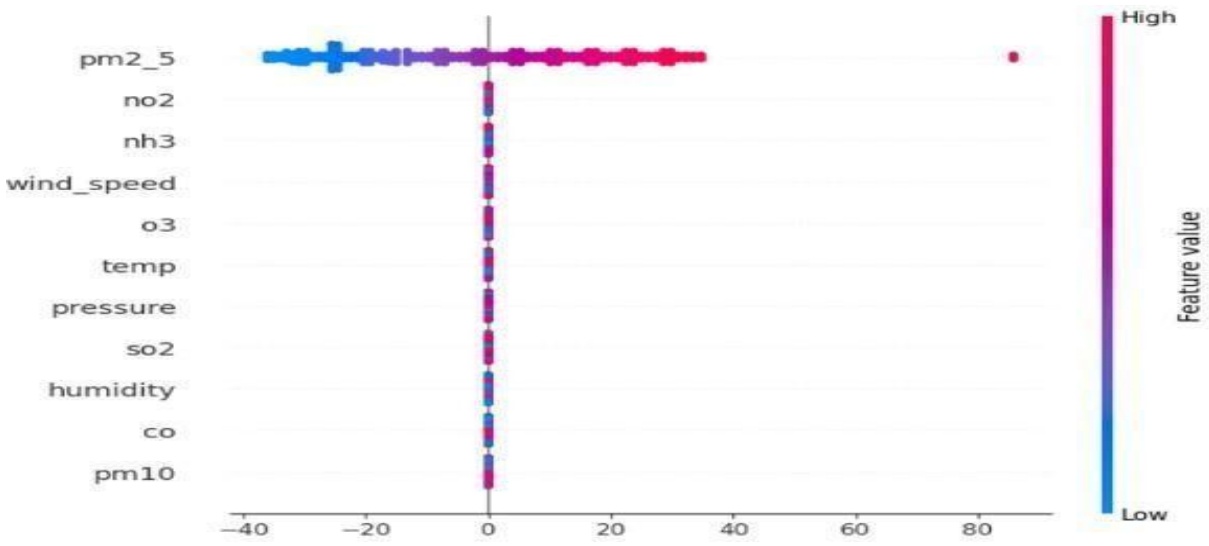
Evaluation Metrics

The value of MAE is 2.62.

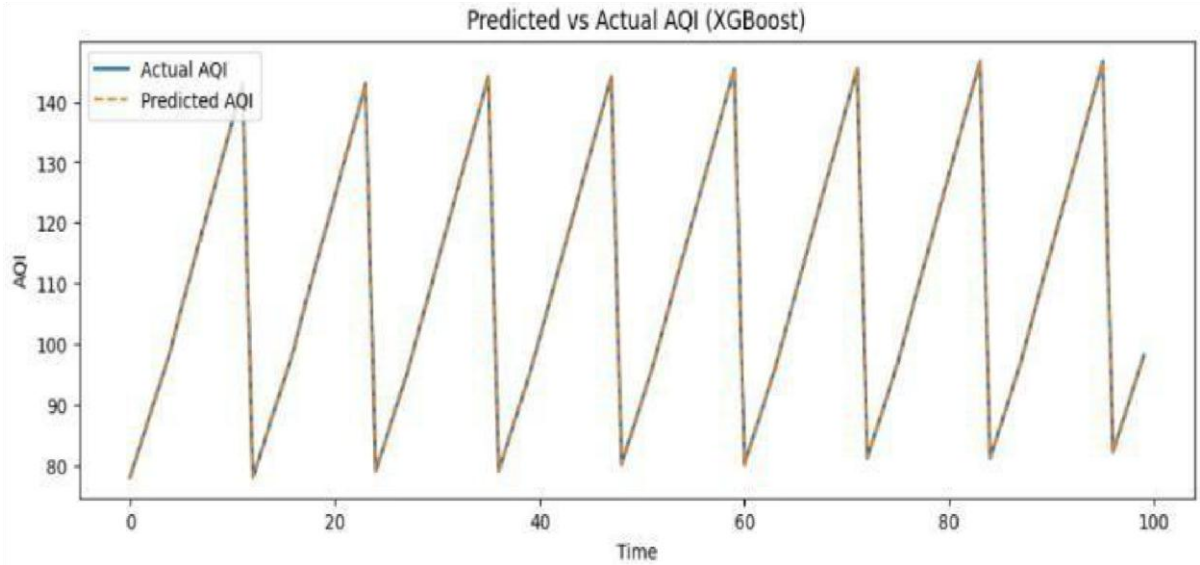
The value of RMSE is 3.27.

The value of R2 score is 0.22.

Using SHAP explainability, it was observed that the feature contributing most to the AQI prediction is PM2.5.



This figure indicates that the model is well-generalized, as all data points closely follow the reference line.



2. RIDGE REGRESSION

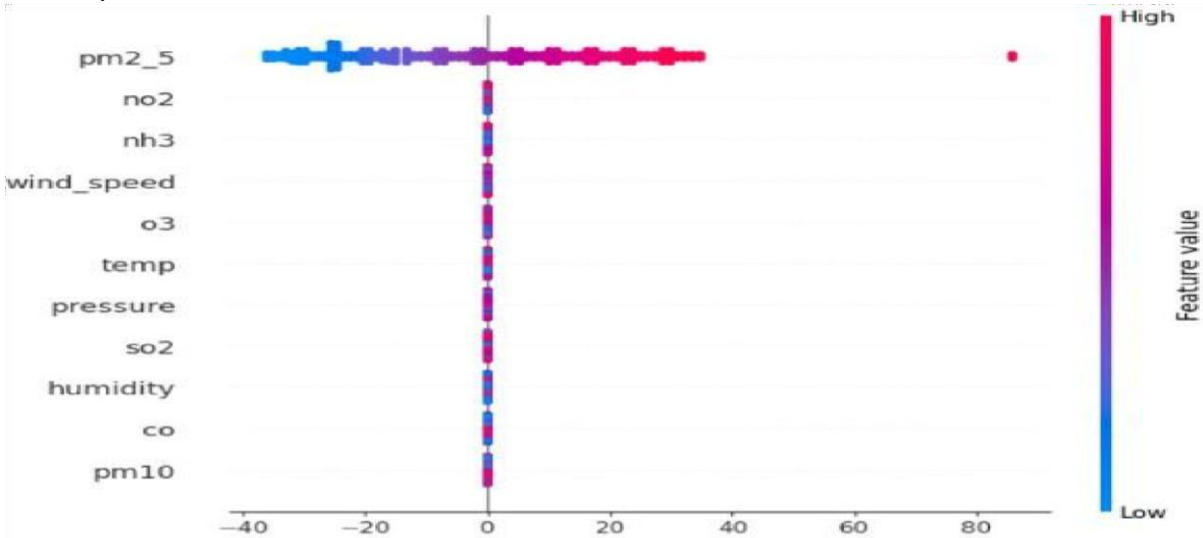
Evaluation Metrics

The value of MAE is 0.54.

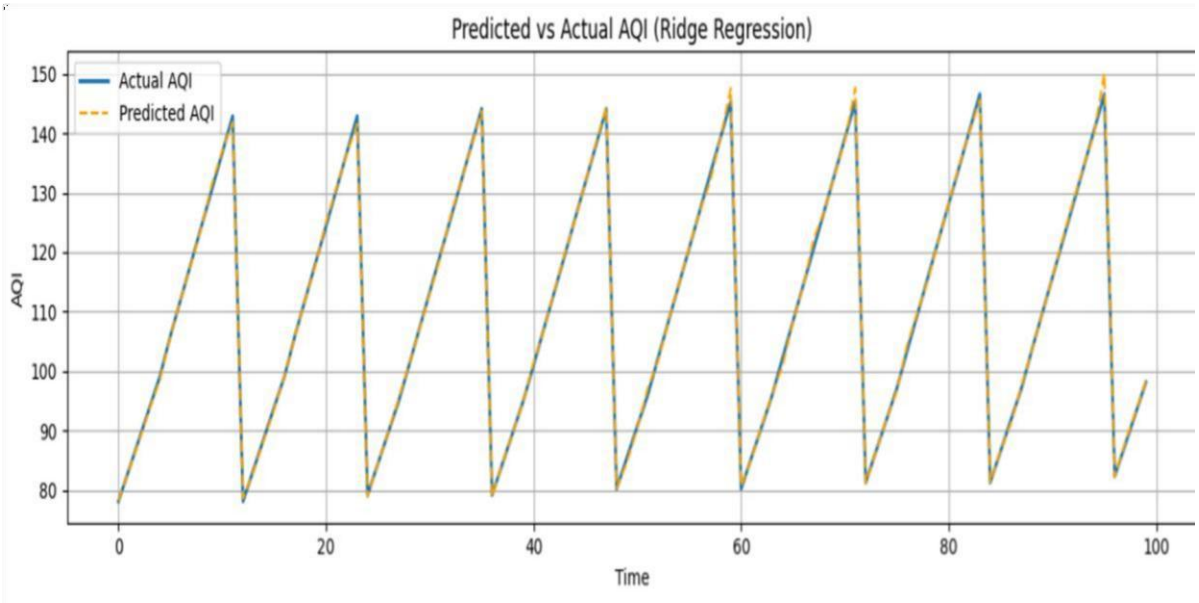
The value of RMSE is 0.75.

The value of R2 score is 0.99.

Using SHAP explainability, it was observed that the feature contributing most to the AQI prediction is PM2.5.



This figure indicates that the model is well-generalized, as all data points closely follow the reference line.



3. LINEAR REGRESSION

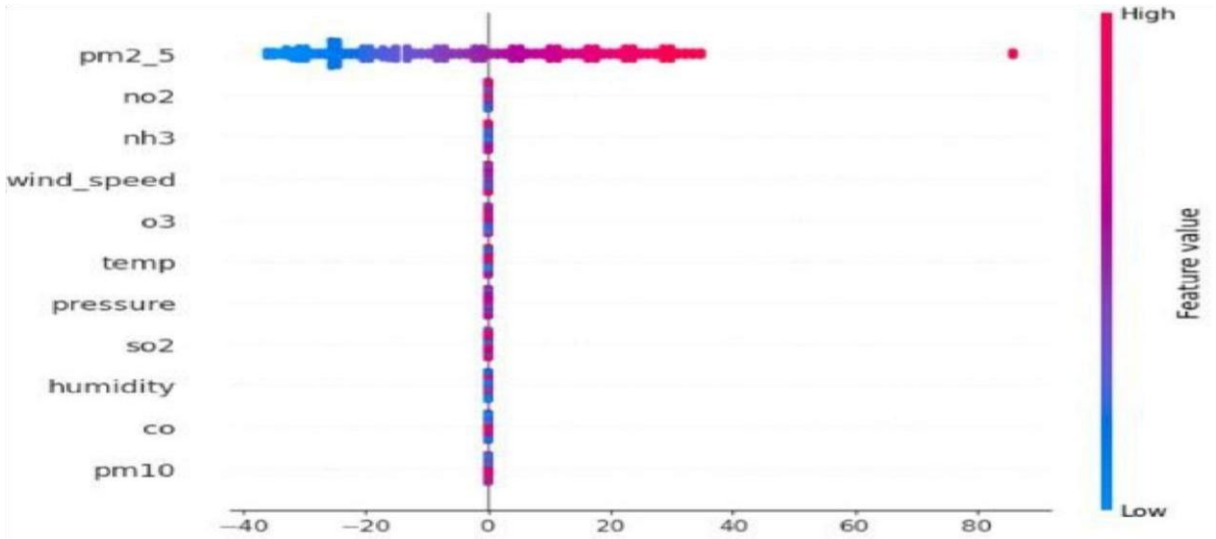
Evaluation Metrics

The value of MAE is 1.87.

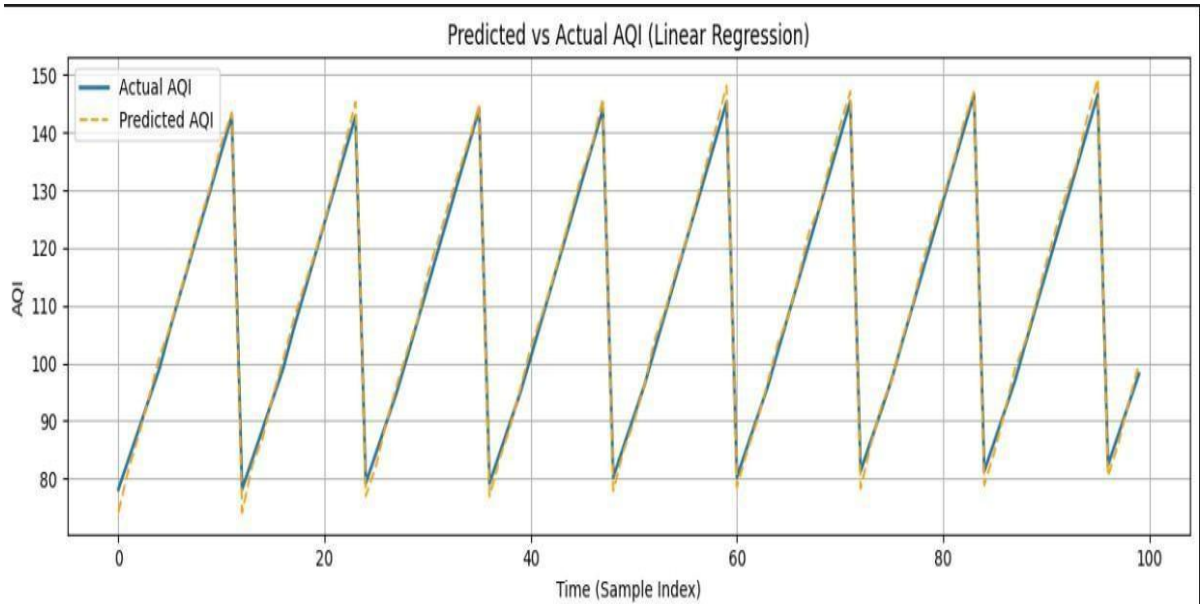
The value of RMSE is 2.38.

The value of R2 score is 0.48.

Using SHAP explainability, it was observed that the feature contributing most to the AQI prediction is PM2.5.



This figure indicates that the model is well-generalized, as all data points closely follow the reference line.



7. Feature Store & Model Registry

A feature store (e.g., Hopsworks or Vertex AI) is used to store preprocessed features for reuse during training and inference. Trained models are saved in a model registry for version control and easy deployment.

8. CI/CD Automation

Continuous Integration and Continuous Deployment (CI/CD) pipelines are set up using GitHub Actions. These automate:

- Hourly: Feature generation from APIs
- Daily: Model retraining with the latest

data

The code is given by

```
name: Hourly AQI Logger

on:
schedule:
- cron: '0 * * * *'  # Every hour   workflow_dispatch:

jobs:
  log-aqi:
    runs-on: ubuntu-latest

    steps:
- name: ✅ Checkout Repo        uses: actions/checkout@v4

- name: 🐍 Set Up Python                          uses:
  actions/setuppython@v4        with:
        python-version:  '3.10'

- name: 📦 Install Python Dependencies          run: pip install
  pandas requests
```

```
- name: Collect and Merge AQI Data run: | python
  - <<EOF import requests import pandas as pd
  import datetime import os
        from glob import glob

        IR_KEY = "a1c4ea3c-c116-43fb-9d98-81004cf3f771"
        WEATHER_KEY = "ee88310481e99a7843bdedc0cca27fd1"

        LAT = 24.8607
        LON = 67.0011
        CITY = "Karachi"
        COUNTRY = "PK"

        def get_iqair_aqi(): try:
              url = f"http://api.airvisual.com/v2/nearest_city?lat={LAT}&lon={LON}&key={IR_KEY}"
r = requests.get(url)
              r.raise_for_status() data = r.json()
pollution = data["data"]["current"]["pollution"] return
{
                "aqi_index": pollution.get("aqius")
              }
            except Exception as e:
              print("IQAir AQI Error:", e)
return None

        def get_openweather_air_pollution(): try:
              url =
f"http://api.openweathermap.org/data/2.5/air_pollution?lat={LAT}&lon={LON}&appid={WEATHER_KEY}"
r = requests.get(url)
              r.raise_for_status() data
= r.json()
              components = data["list"][0]["components"]
return {
                "pm2_5": components.get("pm2_5"),
                "pm10": components.get("pm10"),
                "co": components.get("co"),
                "no2": components.get("no2"),
                "o3": components.get("o3"),
                "so2": components.get("so2"),
                "nh3": components.get("nh3")
              }
```

```
        except Exception as e:
            print("OpenWeather Pollutants Error:", e)
return None

    def get_weather():        try:
            url =
f"http://api.openweathermap.org/data/2.5/weather?q={CITY}
,{COUNTRY}&appid={WEATHER_KEY}&units=metri c"                    r = requests.get(url)
r.raise_for_status()
data = r.json()              return
{
                "temp": data["main"]["temp"],
                "humidity": data["main"]["humidity"],
                "pressure": data["main"]["pressure"],
                "wind_speed": data["wind"]["speed"]
            }
        except Exception as e:
            print("Weather Error:", e)
return None

    def get_forecast_features():        try:
            url =
f"http://api.openweathermap.org/data/2.5/forecast?lat={LAT}&lon={LON}&appid={OPENWEATHER_KEY}&units
=metric"
            r = requests.get(url)
            r.raise_for_status()                    data = r.json()
forecasts = data["list"]            out = {}            for i, entry
in enumerate(forecasts[:24]):
                out[f"forecast_{i+1}_temp"] = entry["main"]["temp"]
out[f"forecast_{i+1}_humidity"] = entry["main"]["humidity"]
out[f"forecast_{i+1}_pressure"] = entry["main"]["pressure"]
out[f"forecast_{i+1}_wind_speed"] = entry["wind"]["speed"]
return out        except Exception as e:
            print("Forecast Error:", e)
return {}

    def collect_hourly_data():
        now = datetime.datetime.utcnow()
date_str = now.strftime("%Y-%m-%d")        time_str
= now.strftime("%H:%M:%S")            hour_of_day =
now.hour                day_of_week = now.weekday()
is_weekend = 1 if day_of_week >= 5 else 0
```

```
    folder = f"data/{date_str}"
os.makedirs(folder, exist_ok=True)    csv_path
= f"{folder}/karachi.csv"

    aqi_data = get_iqair_aqi()    pollutants =
get_openweather_air_pollution()    weather =
get_weather()    forecast = get_forecast_features()

    if not aqi_data or not pollutants or not weather or not forecast:
        print("❌ Incomplete data. Skipping logging.")
return
    row = {
        "date": date_str,
        "time": time_str,
        "hour_of_day": hour_of_day,
        "day_of_week": day_of_week,
        "is_weekend": is_weekend,
        **aqi_data,
        **pollutants,
        **weather,
        **forecast
    }

    df = pd.DataFrame([row])
if os.path.exists(csv_path):
        df.to_csv(csv_path, mode="a", header=False, index=False)
else:
        df.to_csv(csv_path, index=False)

    print(f"✅ Data saved to {csv_path}")

def merge_all_data():
    files = glob("data/*/karachi.csv")
if not files:
        print("No files to merge.")
return
    all_data = pd.concat([pd.read_csv(f) for f in files], ignore_index=True)
os.makedirs("data1", exist_ok=True)
```

```
all_data.to_csv("data1/merged_karachi_aqi.csv", index=False) print("✅ Merged file saved to data1/karachi_aqi.csv") collect_hourly_data() merge_all_data()

EOF

- name: 📂 Commit and Push env:
GITHUB_TOKEN: ${{ secrets.GITHUB_TOKEN }} run: |
git config --global user.name "github-actions" git config --global user.email "actions@github.com" git add data/ data1/ git commit -m "📊 Hourly AQI collected and merged - $(date -u)" || echo "No changes to commit" git push
```

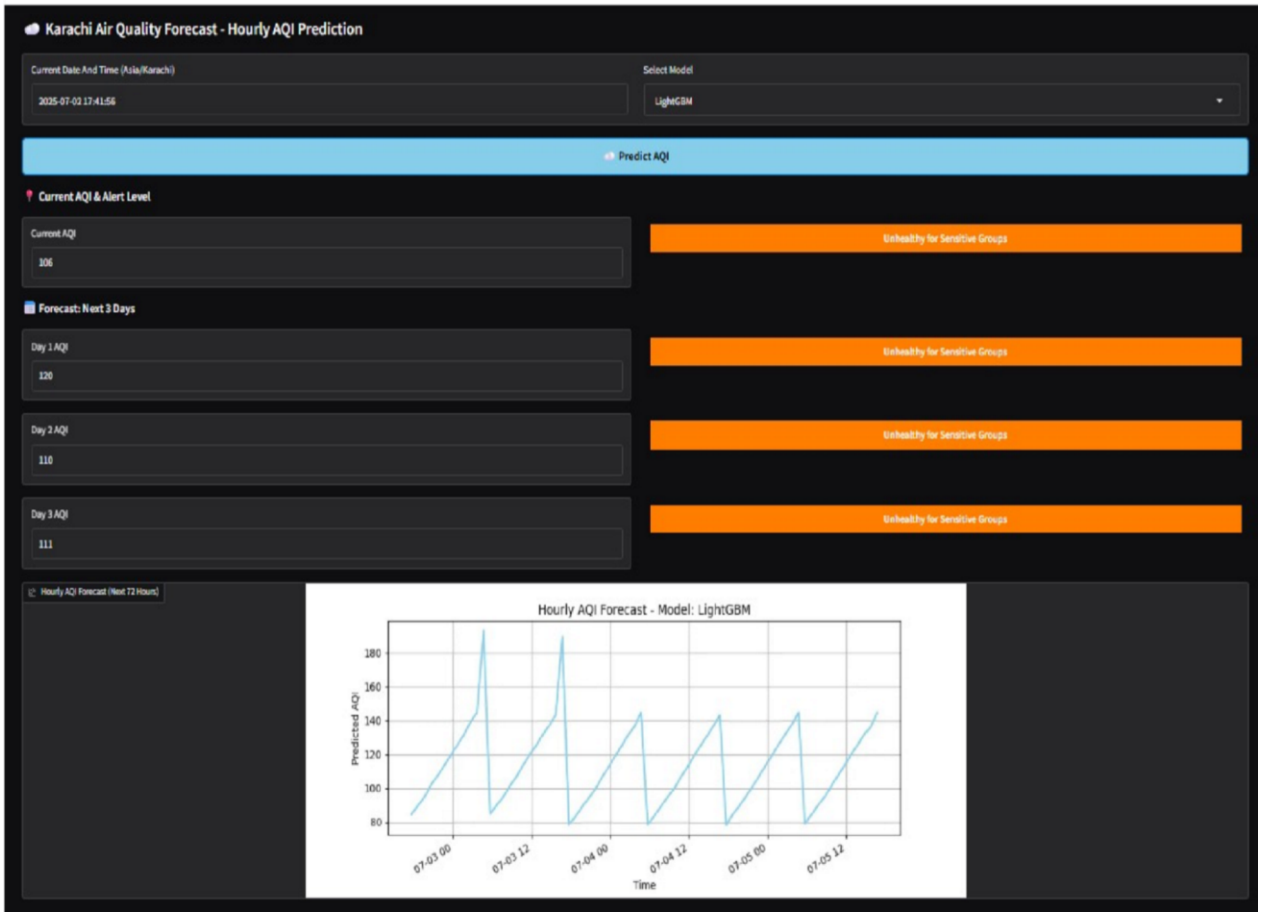
8. API Development

A RESTful API is built using FastAPI to serve real-time AQI predictions. This allows the frontend dashboard and other services to fetch predictions programmatically.

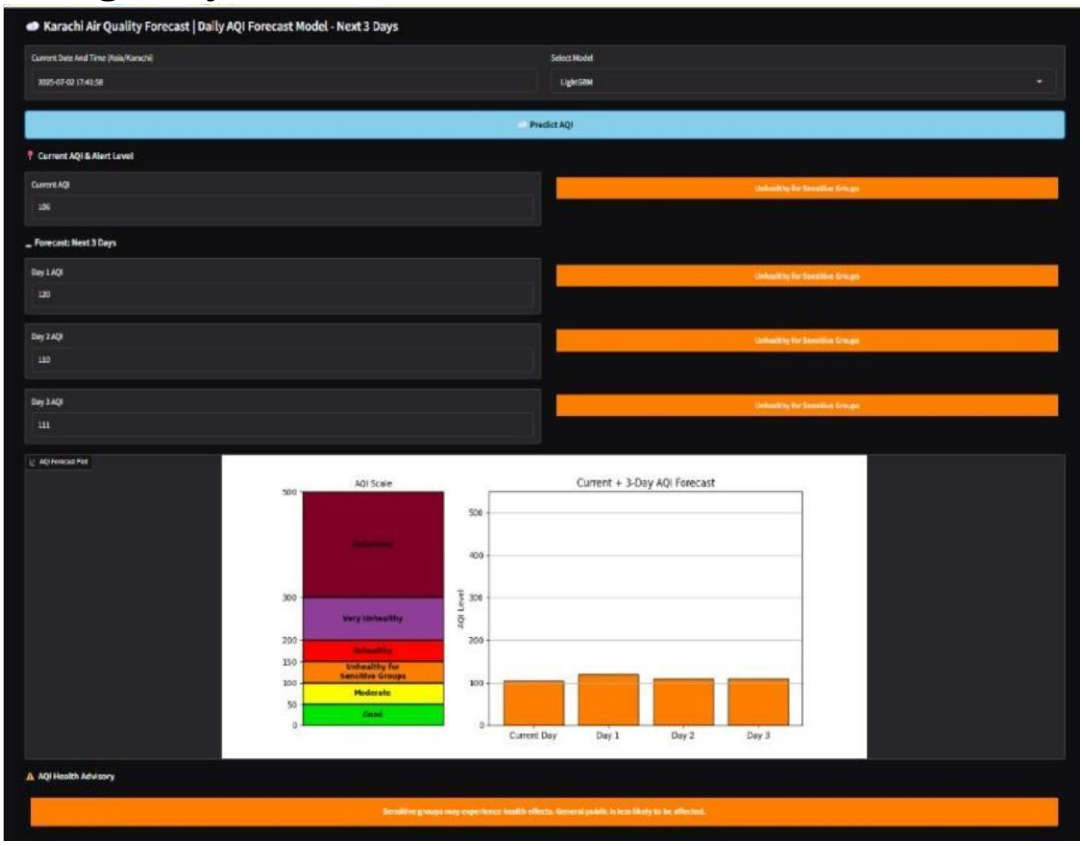
9. Dashboard

An interactive dashboard is developed using Streamlit to display real-time AQI, 3day forecasts, alert levels, and health advisories. It offers users a visual and intuitive interface to monitor air quality.

1. Using Hourly Data Based on Trained Data



2. Using Daily Data Based on Trained Data



2. Real Time Daily AQI Forecast Model – Next 3 Days (Displays current AQI from IQAir, current predicted AQI, and the forecasted AQI for the next 3 days.)